

Region Resonances

Summary

Plain-Language Overview

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Companion documents: Technical Appendix · Methods Primer · Data Appendix · Region Reference — Identity

Abstract. *Region Resonances is a semantic search tool that ranks 59 wine regions by how closely the cultural identity narrative of each region matches a free-form text query. Each region is represented by the concatenation of its one-word identity metaphor and its full identity narrative (typically 280–320 words), embedded once in advance using the OpenAI `text-embedding-3-small` model into a 1536-dimensional unit-norm vector. At query time, the user’s input is embedded by the same model and ranked against the 59 region vectors by cosine similarity (equivalent to the inner product on the unit sphere). The top five matches are presented along with a calibrated display score in $[0, 100]$. Operating characteristics measured on a 60-query calibration set show similarity values concentrated in $[0.15, 0.65]$, with strong matches above 0.50 and the strongest semantically meaningful matches typically below 0.70. The procedure is deterministic given the corpus and the embedding model, and runs in time linear in the corpus size.*

1. Introduction

The Soul of Wine project encodes the cultural character of 59 wine regions through two artefacts: a one-word metaphor (Burgundy is *Devotion*, Mosel is *Poetry*, Santorini is *Survival*) and a roughly 300-word identity narrative that develops the metaphor in context. These artefacts are designed to capture cultural and aesthetic character rather than physical terroir, grape varieties, or production technique.

Region Resonances is a tool for navigating this corpus by query. The user submits a word, phrase, or short prose fragment — for example, *resilience*, *old soul*, or *the feeling of late summer* — and the tool returns the regions whose written identity is most similar to the query in a precise, computable sense.

The aim is not classification or database retrieval. It is closer to what an attentive reader of the project’s prose might suggest if asked which regions a given phrase reminded them of. The tool achieves this by mapping both the corpus and each query into a high-dimensional semantic vector space using a pre-trained language model, then returning the region vectors closest to the query vector.

This document describes the data, the methodology, the empirical operating characteristics, and the robustness of the procedure. The companion *Technical Appendix* provides the formal mathematical treatment; the *Methods Primer* explains the underlying concepts in non-technical terms; and the *Data Appendix* provides the corpus listing and calibration data.

2. Data

2.1 Corpus

The corpus consists of $n = 59$ region passages. Each passage is the concatenation of two artefacts produced by the Soul of Wine project: the one-word metaphor for the region, and the full Layer 1 identity narrative for the region. Passages range from 280 to 320 words. The full set of source narratives is provided in the companion document *Region Reference — Identity*, which is shared between Region Affinities and Region Resonances.

2.2 What the Matcher Does and Does Not Read

The matcher reads only the metaphor and identity narrative. It does not read terroir profiles, grape varieties, vintage notes, scores, or any other conventional descriptor of wine. This scoping is deliberate: Region Resonances is a tool of cultural identity, complementing the Region Affinities tool (which compares identity and terroir) and the TasteRank Explorer (which operates on grape sensory profiles) without overlap.

2.3 Calibration Test Set

A calibration test set of 60 queries was assembled during development to characterise the operating range of the matcher. The set spans single words (e.g., *patience*, *rebellion*), short phrases (e.g., *old soul*, *quiet confidence*), and longer fragments (e.g., *the feeling of late summer*, *a culture that does not need external confirmation*). All queries are in English. The calibration set is not used to train or tune the embedding model — the model is treated as fixed — but to characterise the empirical distribution of similarity scores against this particular corpus.

3. Methodology

3.1 Embedding

Both the 59 region passages and each user query are converted to numerical vectors by the same embedding map: OpenAI’s `text-embedding-3-small` model. The output dimension is $d = 1536$. The model returns unit-normalised vectors; every embedded passage and every embedded query lies on the surface of the unit sphere $S^{d-1} \subset \mathbb{R}^d$.

The 59 region vectors are computed once, in advance, and stored as a static JSON file shipped with the tool. Query vectors are computed on demand at runtime.

3.2 Cosine Similarity

Similarity between a query vector $\mathbf{q}$ and a region vector $\mathbf{r}_i$ is measured by cosine similarity:

$$s_i = \cos(\mathbf{r}_i, \mathbf{q}) = \frac{\mathbf{r}_i \cdot \mathbf{q}}{\|\mathbf{r}_i\|_2 \|\mathbf{q}\|_2}.$$

Because both vectors lie on the unit sphere, the denominator equals one and cosine similarity reduces to the inner product $\mathbf{r}_i \cdot \mathbf{q}$. Computing the full similarity vector $\mathbf{s} \in \mathbb{R}^{59}$ against a query is therefore a single matrix–vector product.

3.3 Ranking and Display

The 59 similarities are sorted in descending order. The top five are presented to the user as the resonance result; the remaining 54 are accessible through a disclosure control. To present scores in a range that uses the full visual scale, the raw cosine values are mapped to a $[0, 100]$ display score via a piecewise-linear clip-and-rescale transformation calibrated on the empirical operating range; details in Section 4.2 and the *Technical Appendix* (Section D).

4. Results

4.1 Worked Examples

The following examples illustrate the kind of result the matcher produces. None of the matches share surface vocabulary with the query; the matching operates at the level of meaning encoded in the embedding space.

Query	Top matches
<i>old soul</i>	Burgundy (<i>Devotion</i>), Rioja (<i>Patience</i>), Tokaj (<i>Melancholy</i>), Sardinia (<i>Stubbornness</i>)
<i>resilience</i>	Santorini (<i>Survival</i>), Barossa Valley (<i>Fortitude</i>), Douro (<i>Endurance</i>), Finger Lakes (<i>Conviction</i>)
<i>the feeling of late summer</i>	Tokaj (<i>Melancholy</i>), Provence (<i>Pleasure</i>), Galicia (<i>Longing</i>)

The matches for *old soul* all describe regions characterised in the corpus by inherited tradition, slow time, and a posture of waiting. The matches for *resilience* cluster around regions whose narratives describe sustained effort against adverse conditions. The matches for *the feeling of late summer* are more impressionistic: a melancholy region, a pleasure region, and a region defined by emotional distance — all plausible interpretations of the seasonal mood.

4.2 Operating Characteristics

Across the 60-query calibration set, observed cosine similarity values s_i fell almost entirely within $[0.15, 0.65]$. Values below 0.15 occur only for pathological inputs (random byte sequences, unrelated languages); values above 0.70 occur only when the query is itself a paraphrase of source-corpus text. This concentration is a generic feature of high-dimensional embeddings of natural-language text: most pairs of English passages share a substantial baseline of common semantic structure (grammar, shared vocabulary, shared world reference), and the signal that distinguishes one passage from another lives in a relatively narrow range above this baseline.

Because raw cosine values in $[0.15, 0.65]$ would be visually compressed and easily misread (a 0.62 looks like a failing percentage but represents a strong match), the display layer applies a clip-and-rescale transformation

$$\tilde{s}_i = \text{clip}\left(\frac{s_i - s_{\min}}{s_{\max} - s_{\min}}, 0, 1\right) \times 100,$$

with $s_{\min} = 0.15$ and $s_{\max} = 0.70$. The transformation is monotonic on the operating range, so it preserves the underlying ranking; it is affine, so equal cosine differences map to equal display differences. A displayed difference of 10 points corresponds to a fixed cosine difference regardless of where on the scale it falls.

4.3 Distribution of Top-Match Scores

On the calibration set, the median top-match cosine across the 60 queries is approximately 0.52 (display score ≈ 67); the median fifth-rank cosine is approximately 0.43 (display score ≈ 51). The typical spread between the top match and the fifth match is therefore around 0.09 in cosine space, comfortably resolvable on the display scale.

5. Robustness Checks

5.1 Determinism

The procedure is fully deterministic given the corpus and the embedding model. Identical queries return identical rankings. The embedding model is treated as fixed and immutable; if the model is updated upstream, the corpus embeddings are regenerated and the change is documented.

5.2 Sensitivity to Query Length

Comparing one-word queries (e.g., *patience*) against multi-sentence queries (e.g., *a culture that values restraint as a moral position*) on the same calibration set shows a small but consistent shift toward higher cosine similarities for longer queries, on the order of 0.05. This reflects the fact that longer queries average over more semantic content and align more closely with the average direction of the corpus. The ordering of matches across the two query forms typically agrees on the top three; rank correlation between the top-10 matches across short and expanded versions of the same query exceeds $\rho = 0.85$.

5.3 Sensitivity to the Embedding Model

A spot-check using OpenAI’s earlier `text-embedding-ada-002` model on a 20-query subset of the calibration set shows top-3 overlap of approximately 70% with `text-embedding-3-small`. The choice of embedding model materially affects rankings at the margin, but the broad structure of which regions resonate with which kinds of query is stable across the two models. Region Resonances commits to a single named model for reproducibility; users who require alternative model behaviour can swap the model with a corresponding regeneration of the corpus embeddings.

5.4 Variation in Narrative Length and Thematic Range

Regions whose narratives interleave several themes (Tokaj, for instance, weaves melancholy, decline, and inherited beauty) tend to appear in the top results for a wider range of queries than regions whose narratives are tightly focused on a single theme. This bias has not been corrected. It is mild and arguably desirable: a region of greater thematic range should resonate with a wider range of inputs.

5.5 Language Sensitivity

The corpus is monolingual (English). The embedding model is multilingual and will produce embeddings for queries in other languages, but those embeddings are systematically biased

toward the model’s training distribution and the rankings are unreliable. Region Resonances should be used with English queries.

6. Conclusion

6.1 Key Findings

Three findings summarise the operation of Region Resonances.

First, semantic embeddings of the Soul of Wine narratives produce intuitively meaningful rankings against free-form queries. The top matches for *old soul*, *resilience*, and *the feeling of late summer* cluster around regions whose narratives describe the corresponding cultural posture, even when no surface vocabulary is shared between query and corpus.

Second, the empirical distribution of cosine similarity values is narrow and asymmetric ($[0.15, 0.65]$ on the calibration set), reflecting the high baseline similarity of natural-language passages in high-dimensional embedding space. Direct presentation of raw cosine values would mislead users; the clip-and-rescale display transformation corrects this without distorting rank order or relative spacing.

Third, the procedure is mathematically well-posed: deterministic, monotonic in similarity, and stable under small query perturbations. Whether the rankings correspond to human judgments of semantic resonance is a property of the embedding model, not of the procedure — a question answered empirically by the calibration set and by user experience.

6.2 Limitations and Future Work

The principal limitation is dependence on the embedding model. The procedure is only as good as the model’s ability to assign nearby vectors to passages a thoughtful reader would judge similar in meaning. A second limitation is the corpus: 59 regions described by single, fixed identity narratives provide a rich but bounded semantic surface. Queries that probe themes outside that surface (e.g., technical winemaking questions, market data questions) will still produce ranked lists but the rankings will not be informative. A third limitation is monolingualism.

Future extensions could include: a multilingual corpus enabling queries in major non-English wine languages; calibration against a panel of expert readers to quantify agreement between embedding similarity and human judgment; and integration with the TasteRank and Region Affinities tools to allow joint queries spanning sensory, terroir, and cultural identity dimensions.

References

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